BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Paper 1: Calculations and Extended Answers

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is a constant?

2. What does an IDE's run-time environment let you do?
   A) Run and test code
   B) Repair physical CPU faults
   C) Rearrange HDD fragments
   D) Manage network user accounts

3. Where are the MSB and LSB in an eight-bit unsigned number?

4. A file stores text, not a photo of a page. "Every letter A is stored as a tiny bitmap." Correct this. How is a letter stored and read?

Today's objectives

Explain the core ideas of paper 1: calculations and extended answers using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Show formulas
Recall and write the calculation rule before substitutions.

Keep units consistent
Convert bits/bytes and MB/GB before arithmetic.

Answer command words
Describe what happens, explain why and evaluate trade-offs.

Earn full tariff
Add linked causal explanations, not repeated versions of one fact.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Show formulas
Recall and write the calculation rule before substitutions.

Keep units consistent
Convert bits/bytes and MB/GB before arithmetic.

Answer command words
Describe what happens, explain why and evaluate trade-offs.

Earn full tariff
Add linked causal explanations, not repeated versions of one fact.

I Do - Worked model

Calculate image size: 120 x 80 pixels at 4-bit colour depth, in bytes.

- Pixels: 120 x 80 = 9600.
- Size: 9600 x 4=38,400 bits.
- Divide eight = 4800 bytes.
- Identify all figures and correct units.

We Do - Guided practice

1. Calculate sound file bits: 8000 Hz, 5 seconds, 8 bits/sample (mono).

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Change the bit depth and predict the new uncompressed raw data size in bytes.

```
width = 12
height = 10
bitsPerPixel = 4
bytes = width * height * bitsPerPixel DIV 8
print(bytes)
```

You Do - Independent exam questions

Q2. Calculate 2560 bits in bytes. [2 marks]

Q3. Convert 3 MB to kB under decimal units. [2 marks]

Q4. Explain why two distinct SSD benefits may score better than repeating 'fast'. [3 marks]

Exit Ticket - True or False?

When calculating sound size, sampling rate can be ignored.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. A named value that stays the same while the program runs.

2. A - Run and test code

3. The MSB is the leftmost bit, weight 128; the LSB is the rightmost bit, weight 1.

4. Each character has a number code stored in binary. Both computers need a matching encoding to show the intended characters.

I Do - Explain the model

Pixels: 120 x 80 = 9600.

Size: 9600 x 4=38,400 bits.

Divide eight = 4800 bytes.

Identify all figures and correct units.

We Do - Model answers

1. No header/compression assumptions.

Code First - Answer

```
width = 12
height = 10
bitsPerPixel = 8
bytes = width * height * bitsPerPixel DIV 8
print(bytes)
```

Expected output: 120

You Do - Self-mark

Q2: 2560 / 8 = 320 bytes.

Q3: 3000 kB.

Q4: Specific benefits (e.g. impact durability and access time) address different marking points.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.